From Topic to Talk in One Pipeline

A 45-minute practical with five AI research tools

LOURIM, AI for Research Seminar Series, Part 1

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What you will do today. You received a research topic on a slip of paper, randomly assigned by a peer. By the end of this session you will have *started* a pipeline that ends, at home, with two deliverables:

- a short **document** that gives an overview of recent progress in the field and lays out a few promising research questions attached to it;
- a **slide deck** built from that document, which you can reuse to pitch the questions and find potential collaborators.

You will email both to a researcher working in the field, asking which of those questions might be worth pursuing. The five tools below each take care of one step.

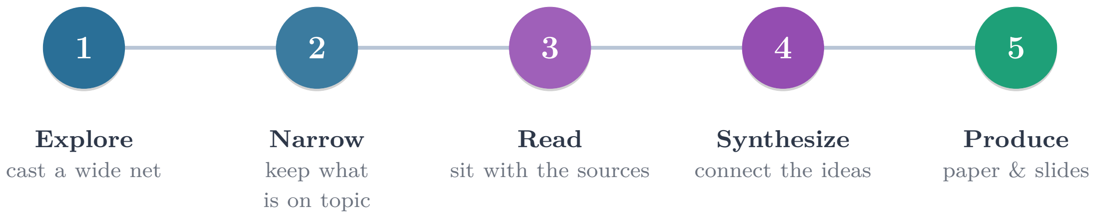
Scan to open the source repo.
All slides, this tutorial, and the upcoming seminar materials live at github.com/cvandekerckh/lourim-ai-seminar-series.

Contents

1 Overview

1.1 From topic to talk: the method

Before we name any tool, here is the recipe behind doing good research, with or without AI. Five steps, in order:



Explore. You start from a topic, not a question. Look broadly at what already exists, even outside your own field, so you do not miss the literature that matters.

Narrow. Filter the long initial list into a small, trustworthy set you can actually engage with. Drop the off-topic and the weak.

Read. Spend real time with each source. Note the question it asks, the method it uses, what it finds, and where it is fragile.

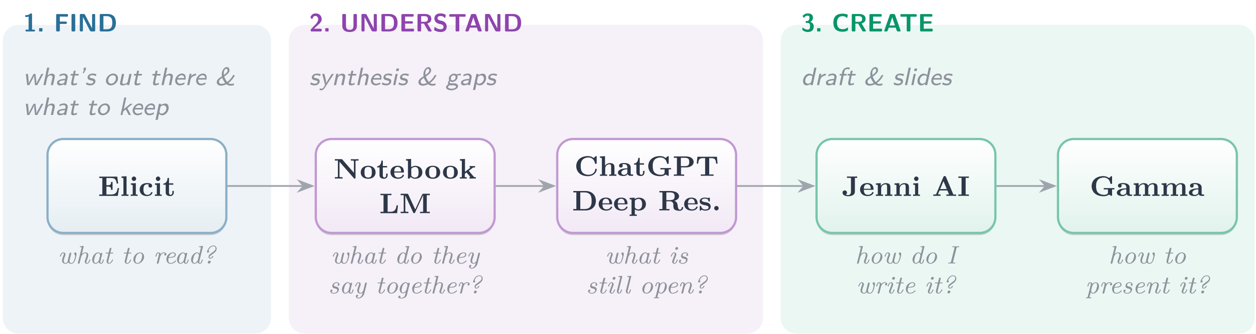
Synthesize. Step out of individual papers and look at the picture they make together: what is the consensus, where do they disagree, what is missing?

Produce. Turn that synthesis into something other people can engage with: a written argument for the open research questions you found, and a presentation that does it justice.

This recipe is the work of any researcher. The five AI tools below are nothing more than *assistants* for each step. They speed up the recipe, they do not replace it.

1.2 The five tools

Each tool takes care of one step of the recipe and answers one specific question:



1.3 Time budget

The 45-minute session is a **kickoff**, not the whole journey. The plan below is a guideline. The real instruction is simple:

Tip

Go as far as you can in 45 minutes. Some of you will finish Elicit only, some will reach NotebookLM, and that is fine. The rest is guided homework you finish over the following days.

What matters is that you start the pipeline today and that you keep moving forward at your own pace.

Phase	What happens	Suggested time
Intro	Recap pipeline; check accounts; pick your topic phrasing	5 min
§ 3 Elicit	Search, filter, build the table, download 3 PDFs	20 min
§ 4 NotebookLM (start)	Upload the papers, run the synthesis questions	15 min
Wrap-up	Brief on what to do at home	5 min
Homework	At your own pace, around 3 to 5 h spread over a week	

If you fall behind in the room, do *not* skip steps to catch up. Stop where you are, take a clean note of where you stopped, and pick up from there at home. Skipping steps is what produces a paper you cannot defend.

1.4 Golden rule

Tip

You are the editor; the AI drafts. Every output of every tool is a starting point you read, prune, and correct. If you cannot defend a sentence to a colleague, delete it.

2 Before you start

1. **Topic in hand.** The slip of paper a peer gave you. If it feels too broad (e.g. "leadership"), narrow it to one question (e.g. "how does remote work change leadership in teams?").

2. **A browser open** with you signed in to the five tools. The free versions are enough; sign-up links are in Appendix A.

3. **A folder for this project**, somewhere you will find it again, on your computer, in Dropbox, in Google Drive, wherever you usually work. Inside it, make four sub-folders: *Papers*, *Notes*, *Draft*, *Slides*. You will save things there as you go.

4. **A document to keep your prompts.** Open a Word document (or a Google Doc, or Notes) and call it "My prompts". Each time you ask one of the tools a question that worked well, paste the question in. Later, this lets you remember what you did and explain it if asked.

5. **Do not upload anything sensitive or unpublished**, your own work, your colleagues' work, or interviews you collected. Stick to published papers and to notes you are happy to share.

3 Elicit: search and screen

Goal. Find the 3 papers most relevant to your topic, and a quick read of the field around them.
Input. Just your topic in a few words.
Output. A comparison table you can read in Elicit (research question, main finding, year), plus 3 downloaded PDFs in your *Papers* folder.

Sign up

Go to elicit.com and click "Sign up". The free version is enough for this exercise.

Steps

1. In Elicit, choose the *Find papers* workflow. Type your topic phrased as a research question (for example, "how does remote work change leadership in teams?"). Elicit will return a ranked list of papers.
2. Apply three filters before you look at any abstract:
   - *Has PDF available*
   - *Q1 journals only*
   - *Published in the last 6 years*
3. Add three columns to the table: *Research question*, *Main finding*, *Year*.
4. Click "Run" so Elicit fills the cells. Read them. If something looks wrong, it probably is. Elicit sometimes summarizes too confidently.
5. Exporting the table is not available on the free plan, so pick the 3 papers that look most relevant to you, google their titles, and download the PDFs into your *Papers* folder.

Watch out

Elicit can invent. The cells are summaries written by an AI, not quotes from the paper. Before you put a number, a sample size, or a finding into your own paper, open the PDF and check it with your own eyes.

Time budget: 20 min.

4 NotebookLM: synthesize your corpus

Goal. Understand, as a whole, what your 3 papers are saying together.
Input. The 3 PDFs you downloaded.
Output. A set of notes about the main themes, agreements, disagreements, and open questions.

Sign up

Go to notebooklm.google.com and sign in with a Google account. It is free.

Steps

1. Click *New notebook* and give it the name of your topic.
2. Upload the 3 PDFs you downloaded as "sources".
3. On the right side there is a chat box. Paste the four questions from the *Copy-paste prompt* box just below, one by one. After each answer, click *Save to note* so it lands in your Notes panel.
4. Once all four answers are saved, open the note in NotebookLM and use the *Export to Google Docs* option. Move the resulting Doc into your *Notes* folder.

Copy-paste prompt

```
1. What are the 4 to 6 main themes that recur across these sources?
2. Where do the sources agree, and where do they disagree? Cite the sources.
3. Which definitions of the central concepts vary across the sources, and how?
4. What questions do these papers leave unanswered?
```

Watch out

A citation does not mean "correct". NotebookLM points you to the file it used, but its summary of that file can still be wrong. Click each little citation number to jump to the source and check.

Time budget: 15 min in session to upload and run prompts; finish curation at home.

5 ChatGPT Deep Research: find the gaps

Goal. Step back from your papers and ask: what are the open questions in this field?
Input. The notes you saved from NotebookLM.
Output. A short "insights" document you will use to write the paper.

Sign up

Go to chatgpt.com. Deep Research is only available on the paid plan. If you do not have it, the same prompt also works in regular ChatGPT with the web search option turned on.

Steps

1. In ChatGPT, click the "+" button on the left of the message box and choose *Deep Research* (or *Recherche approfondie* in French). If you do not see it (free plan), skip this step and just paste the prompt into a regular ChatGPT chat; the result is still useful.

2. Copy the prompt below into the message box. Replace `<TOPIC>` with your topic, and paste your NotebookLM notes underneath where it says "NOTES".

3. Send it. Deep Research takes 5 to 15 minutes; this is normal. You can leave the tab open and do something else.

4. When the report comes back, read it slowly. For each claim you want to keep, click the source link. If the link is broken, or if the source does not actually say what is claimed, drop the claim.

5. Download the .docx that ChatGPT produces (the prompt asks for one). Open the NotebookLM Google Doc you exported earlier, and copy-paste the cleaned gaps content at the end. You now have a single Google Doc with two parts: the state-of-the-art overview from NotebookLM, then the open research gaps from Deep Research.

Copy-paste prompt

```
You are a research assistant for a literature-review draft on <TOPIC>.
Below are my synthesized notes from a small curated corpus.  Using these notes plus
the broader literature, produce:
1.  The 3 to 5 most consequential research gaps.
2.  The main methodological challenges in the field.
3.  A side-by-side comparison of the dominant theoretical perspectives.
4.  Two or three concrete, testable directions for future work.
For every claim, cite at least one source with a working link.  Flag uncertainty
explicitly.
At the end, package the answer as a downloadable .docx file I can grab.
=== NOTES ===
<paste NotebookLM notes here>
```

Watch out

Made-up citations. Even Deep Research sometimes invents references that look real. Open every link before you trust a claim. If the link is dead, or the page does not say what was claimed, throw the claim out.

Time budget: 15 min active plus waiting time. Homework.

6 Jenni AI: draft the paper —optional

Why Jenni. Jenni shines when you (a) need help *starting from a blank page* and want a guided path from topic to outline to paragraphs; (b) want citations attached as you write, with reference checking against your uploaded Library; (c) want a clean, consistent bibliography style applied automatically. If you already know how to draft and just need polish, you can skip this tool.
Purpose. Use Jenni AI to write the paper progressively, one selected paragraph at a time, with the merged notes always at hand as a citable source.
Goal. Draft a short document (about 1 500 to 2 000 words) that argues for the open research questions you identified.
Input. The merged Google Doc (state-of-the-art overview + Deep Research gaps), exported as PDF.
Output. A first draft, organized in sections, with the open questions and where to go next as its centre of gravity.

Watch out

Pricing limits what you can do here. Jenni's free tier is small, and full use of *Modify with AI* + Library citations sits behind the paid plan. The flow below is the ideal experience and gives you the idea. If the free quota runs out, you can fall back on (a) good prompting on a free GenAI website such as ChatGPT or Claude, pasting paragraphs of your draft and asking for the same rewrite, or (b) skipping straight to the Gamma section and feeding it your merged notes directly to build the slides.

Sign up

Go to jenni.ai. The free version has a small daily word limit, enough to write one short paper.

Steps

1. Open the merged Google Doc and export it: *File → Download → PDF Document*. Save the PDF in your *Draft* folder, then upload it into Jenni's *Library* so the AI can cite from it.
2. In Jenni, click *New document*. Skip any startup prompt and start writing straight away.
3. Type your topic on the first line.
4. Double-click on the topic to select it, choose *Modify with AI*, toggle the source to *Library* (not *Web*), and send *Step 1* of the prompt box below. Click *Replace selection* to drop the answer into the document.
5. Select all the text (Cmd/Ctrl-A), open *Modify with AI* again with *Library* on, send *Step 2*, and click *Replace selection*. The document now holds an outline with one paragraph per section.
6. Select the introduction paragraph, open *Modify with AI* with *Library* on, send *Step 3* after filling in three themes specific to your topic, and click *Replace selection*.
7. Repeat the previous step for the methodology, discussion, and conclusion paragraphs, adapting the three themes each time so they fit the section.
8. As you go, use Jenni's *Add citation* button on sentences that need a reference; it will surface candidate papers from your Library and let you insert them inline.
9. Save the final paper as a Word document in your *Draft* folder, under the name "Paper, version 1".

Copy-paste prompt

```
Step 1 - Identify the paper idea
Here is a contextual document for my research paper.  Identify:
1.  the main research topic
2.  the possible research contribution
3.  an appropriate academic paper structure

Step 2 - Generate the outline
Create a detailed research paper outline based on this document.

Step 3 - Rewrite each section using the library notes (used per section)
Rewrite this introduction using the uploaded notes.
Focus specifically on:
- <THEME 1 of your topic>
- <THEME 2 of your topic>
- <THEME 3 of your topic>
Avoid generic statements.  Use academic style.

For methodology, discussion, and conclusion:  replace "introduction" with the section
name and adapt the three themes to that section.
```

Tip

After each *Modify with AI* answer, click *Replace selection* so the new text lands in the document. The paper grows paragraph by paragraph, never as one big dump.

Watch out

Wrong citations. Jenni may suggest a citation that looks perfect but does not actually support the sentence. Open the paper and check before you keep it.

Time budget: 60 to 90 min. Homework.

7 Gamma: generate the slide deck

Goal. Turn your work so far into a slide deck of 7 to 10 slides.
Input. A single file: either your merged notes (from Deep Research and/or NotebookLM), or your final paper (the Jenni AI output, if you did the optional step).
Output. A PowerPoint (.pptx) deck you can keep editing afterwards.

Sign up

Go to gamma.app. The free version has enough credits for one deck.

Steps

1. Click *Create*.
2. Choose *Import a file*, then *Upload a file*.
3. Upload one of: your merged notes (Deep Research + NotebookLM) *or* your paper from Jenni AI if you did the optional step.
4. When asked what to generate, choose *Presentation*.
5. Set the configuration: pick a theme, set text quantity, and toggle AI image generation if you want generated visuals. For the artistic style, choose *Photo* (it produces the most credible, least cartoonish visuals for an academic audience).
6. Click *Generate*. After a moment, the deck opens.
7. Edit the first slide so it shows your name, your topic, the date, and LOURIM. Add or remove slides so the order tells the story you want.
8. If Gamma added an image you cannot defend, replace it or delete it. When in doubt, no image is better than the wrong image.
9. Click *Export* → *PowerPoint* (.pptx). Save it in your *Slides* folder under the name "Slides". Open it in PowerPoint or Keynote later to refine.

Tip

You can run Gamma in parallel with Jenni: while Jenni is generating a section, switch tabs and start Gamma on the parts you have so far.

Watch out

Generated charts are decoration, not data. If Gamma adds a chart, it is invented from the words of your paper, not from real numbers. Never present it as if it were a real measurement.

Time budget: 30 min. Homework.

8 Send to the researcher

The last step, about a week after the session, is to send your work to a researcher who works on your topic.

Choose the person

Pick the researcher you (or the organizer) identified for your topic. The simplest choice is the author who comes up most often in the papers you kept, and whose university email address is easy to find online. One person is enough.

What to attach

- Your final paper (PDF) or, if you skipped Jenni, the merged notes (PDF).
- Your slide deck. Open the PowerPoint in PowerPoint or Keynote, refine it, then export as PDF for the email.

Email template

Copy-paste prompt

```
Subject: Open research questions on <TOPIC>, LOURIM AI seminar (worth pursuing?)

Dear Prof. <NAME>,

As part of an AI for research seminar at LOURIM (UCLouvain), I put together a short
document mapping what I see as the open research questions on <TOPIC>, drawing
notably on your work on <SPECIFIC PAPER>.

The document and a short slide deck are attached. The exercise was time-boxed
and AI-assisted, so it is exploratory. I would value your view on which of those
questions, if any, look worth pursuing.

With thanks for your time,
<YOUR NAME>
LOURIM, UCLouvain
```

Watch out

Disclose AI use. The email already says "AI-assisted"; keep that line. Researchers appreciate transparency far more than they mind the use of tools.

9 Risks: a six-line reminder

Hallucinations	Check every fact, number, and citation against the original source.
Bias	These tools reflect what they were trained on. Be aware whose voices may be missing.
Confidentiality	Never upload unpublished data, sensitive interviews, or internal LOURIM documents.
Traceability	Keep your "My prompts" document up to date. You will be glad later.
Scientific integrity	You have to be able to defend each argument as your own.
Responsibility	It is your name on the paper, not the tool's.

A Accounts and free tiers

Tool	Sign-up	Free-tier note
Elicit	elicit.com	Free monthly credits, enough for one screening.
NotebookLM	notebooklm.google.com	Free with a Google account.
ChatGPT Deep Res.	chatgpt.com	Paid plan; fallback: regular ChatGPT with web search on.
Jenni AI	jenni.ai	Free daily word cap; full *Modify with AI* + Library is paid. Fallback: ChatGPT/Claude.
Gamma	gamma.app	Free credits cover one deck.

B Troubleshooting

- **Elicit returns very few papers after filtering.** Loosen one filter at a time: try the last 8 years instead of 6, or drop the Q1 restriction. Re-check that your topic is phrased as a question, not a single keyword.
- **Elicit ran out of free credits.** Run the extraction only on the 3 papers you plan to keep, not on the full ranked list.
- **NotebookLM refuses to upload a PDF.** The file is probably too big or it is a scanned image. Split the PDF, or use a free tool like Adobe to turn the scan into real text.
- **No access to Deep Research.** Paste the same prompt into the regular ChatGPT with the web search option turned on; the result is shorter but still useful.
- **Jenni adds citations you do not recognize.** Toggle the source to *Library* (not *Web*) so it only cites the PDF you uploaded.
- **Jenni's free quota is exhausted.** Stop there and use ChatGPT or Claude with the same prompts (paste your draft as context), or skip ahead to Gamma using your merged notes as input.
- **Gamma exports an ugly PDF.** Use the built-in *Export to PDF* button, not the browser's "Print" function.

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